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RESEARCH ARTICLE

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Artificial intelligence-based algorithm for cervical vertebrae maturation stage assessment

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Abstract

Objective: The aim of this study was to develop an artificial intelligence (AI) algorithm to automatically and accurately determine the stage of cervical vertebra maturation (CVM) with the main purpose being to eliminate the human error factor.

Setting and Sample Population: Archives of the cephalometric images were reviewed and the data of 1501 subjects with fully visible cervical vertebras were included in this retrospective study.

Materials and Methods: Lateral cephalometric (LC) that met the inclusion criteria were used in the training process, labeling was carried out using a computer vision annotation tool (CVAT), tracing was done by an experienced orthodontist as a gold standard and, in order to limit the effect of the uneven distribution of the training data set, maturation stage was classified with a modified Bachetti method by the operator who labelled them. The labelled data were split randomly into a training set (80%), a testing set (10%) and an validation set (10%), to measure intra-observer, inter-observer reliability, intraclass correlation coefficient (ICC) and weighted Cohen's kappa test was carried out.

Results: The ICC was valued at 0.973, weighted Cohen's kappa standard error was 0.870 ± 0.027 which shows high reliability of the observers and excellent level of agreement between them, the segmentation network achieved a global accuracy of 0.99 and the average dice score overall images was 0.93. The classification network achieved an accuracy of 0.802, class sensitivity of (pre-pubertal 0.78; pubertal 0.45; post-pubertal 0.98), respectively, per class specificity of (pre-pubertal 0.94; pubertal 0.94; post-pubertal 0.75), respectively.

Conclusion: The developed algorithm showed the ability to determine the cervical vertebrae maturation stage which might aid in a faster diagnosis process by eliminating human intervention, which might lead to wrong decision-making procedures that might affect the outcome of the treatment plan. The developed algorithm proved reliable in determining the pre-pubertal and post-pubertal growth stages with high accuracy.

KEYWORDS
artificial intelligence, cervical vertebrae, deep learning, orthodontics

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1 | INTRODUCTION

Timing in orthodontics and dentofacial orthopaedics is one of the decisive factors for successful treatment. Growth modification will depend mainly on choosing the exact period of time to intervene, and the peak of growth (growth spurt) is considered to be the best time for intervention to utilize the remaining growth period to correct any skeletal problems.[1,2]

A growth spurt can be identified as a period of accelerated or intense physiologic growth and determining the beginning of this period of growth can be a challenge; however, utilizing the growth spurt period will guarantee a successful growth modification.[3,4]

Several assessment methods, of which the most accurate proved to be hand and wrist x-rays which in itself have the disadvantage of subjecting the patient to an extra dose of radiation, have been described over the years in an attempt to predict growth spurt.[3]

The cervical vertebra seen on a routine LC has been used to determine skeletal maturity. It is known that the lateral view of cervical vertebral bodies changes with continuous growth.[4-6]

The Bachetti method implemented in this study described six definite and consecutive stages of assessment depending on the shapes of the cervical vertebrae C2, C3 and C4, all of which correspond to the peak mandibular growth.[1]

Artificial intelligence (AI) applications can assist orthodontists in making better decisions and obtaining better results as their measurements are highly accurate, and their implementation helps prevent possible human-made errors.[7-11]

Deep learning can be defined as the process of training a computer to learn like a human brain.[12,13]

Deep learning involves multilayer learning patterns using artificial neural networks (ANNs). An artificial neuron is a type of cell that obtains input, makes the necessary calculations and yields the results just as a biological neuron does.[13,14]

Previously conducted studies regarding AI-assisted cervical vertebra maturation (CVM) stage determination are limited because implementing AI is a fairly recent practice in orthodontics.

Kok et al[15] tested seven different algorithms usually used for classification on a sample of 300 subjects in which they placed the reference points and conducted linear measurements. The ANN showed high accuracy in their study, encouraging us to utilize ANN with a different technique and larger sample.

An algorithm to automatically stage CVM was developed in a study carried out by Amasya et al[16] in 2020. Similar to the study by Kok et al,[15] a total of 647 LC were used and the reference points and linear measurements determined the morphological features of each vertebra.

Using a similar technique, Zhou et al[17] opted to use a sample of 1080 subjects. Although this sample size is an improvement, the drawback of this study is its distribution, which is limited to only the training and testing groups. So, the inclusion of a dataset of a validation group could help tune the algorithm and help it reach maximum ability.

The previously mentioned three studies all opted for a supervised learning AI model. Seo et al[18] are the first to use an unsupervised learning AI model in CVM staging; however, they did not use linear measurement or reference points. Instead, they used a gradient-weighted class activation map (Grad-CAM) to teach the vertebras the changes.

To overcome the drawbacks of the previously mentioned studies, we collected a larger sample and divided it into three groups which contained a validation set to tune the algorithm. A convolutional neural network (CNN) was implemented for image recognition, and instead of using reference points and linear measurements, we opted for tracing the vertebra to apply unsupervised learning in our model.

This study is the first to implement the CNN model and unsupervised learning to automate the CVM staging process and also is the first to implement tracing for the AI to learn the vertebras as shapes. Another point worth mentioning is that the present study has the largest sample size compared to all the previous studies.

2 | MATERIALS AND METHODS

In accordance with the regulations of the 1964 Helsinki Declaration on medical research ethics, a request was submitted to and approved by the Near East University Scientific Research Evaluation Ethics Committee (decision date and number: YDU/2021/93-1382).

2.1 | Preparation of dataset

Data were collected from the Orthodontic Department Archive at Near East University (2015-2021). All the LCs were taken by Orthophos XG (Sirona Dental Systems GmbH). The collected data constituted a wide range of age groups (7-25 years) in order to ensure the evaluation of all maturation stages of CVM.

The exclusion criteria were:

- Any presence of an artefact (Beam-hardening artefact, motion artefact).
- Images without the full body of C2 C3 C4 in the field of view.
- Presence of any syndrome that would affect the maxillofacial structures.
- Improper position of the patient's head.

In total, 1501 LCs in total qualified to become part of the research and all LC cases were pseudonymized before proceeding to tracing and labelling.

After examining the collected data, we noticed a major difference in sample sizes between the different maturation stages. Thus, to reduce the effects of this limitation, we simplified the classification procedure and merged two sequential classes into a single class. In brief, the stages of CVM were divided into three categories:

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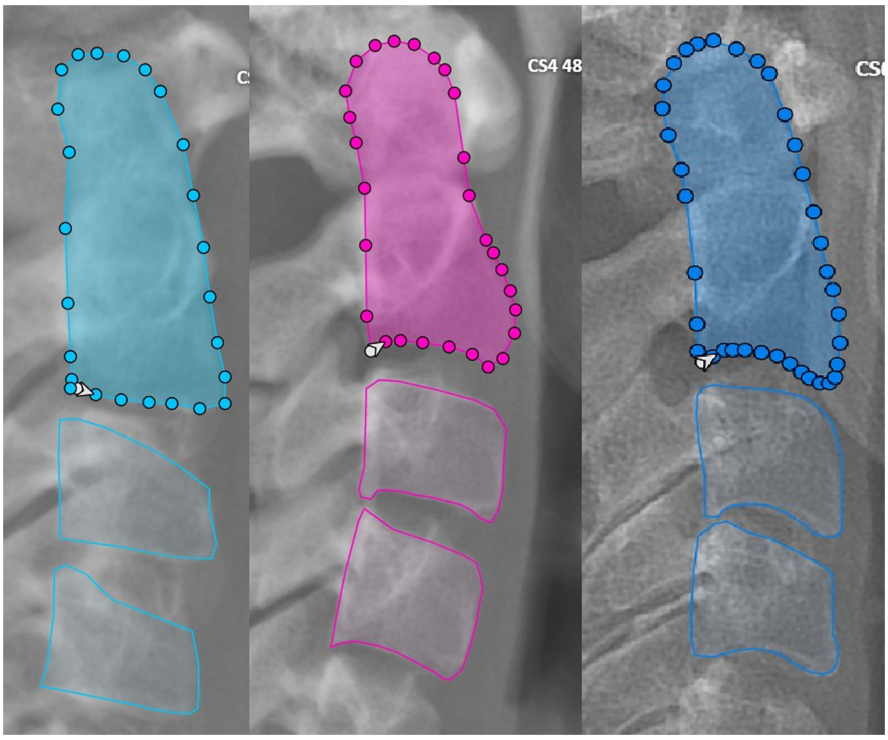


FIGURE 1 Tracing process of vertebras, each distinct colour represents a different stage of CVM.

1. Pre-pubertal stage (CVM1 and CVM2).
2. Pubertal stage (CVM3 and CVM4).
3. Post-pubertal stage (CVM5 and CVM 6).

The principal observer (orthodontic resident) traced the vertebras using the computer vision annotation tool (CVAT) and assigned a class for the trace. This tracing was used to segment the vertebras in the initial phase, and the class for the trace was used for classification in the second phase (Figure 1).

To evaluate the reliability, the principal observer re-classified 20% of the data separately 2 weeks after they finished the first annotation. In addition, the same data were given to a second observer for re-classification to measure the reliability between the two orthodontists. The computer randomly generated the selected data used for the re-evaluation.

When the preparation of the dataset was completed, the labelled data were randomly split into a training set (80%), a testing set (10%) and a validation set (10%).

2.2 | Model pipeline

Our study approach contained the following steps:

1. Each LC X-ray was resized to 1024 by 768.
2. Using U-Net architecture each LC X-ray was segmented.
3. From each segmented LC the C2 C3 C4 was isolated.
4. A patch centred on the vertebras was cropped.
5. The data were then run through a classification network based on Alex-Net that resulted in the classification of CVM stages. (Figure 2).

2.3 | U-Net

U-Net is an algorithm whose main focus is on image segmentation. It is based on CNNs and is designed to segment biomedical images in which the input is an image and the output would be another related image.[19]

U-Net architecture gets its name from its shape. It has a U-shaped symmetric architecture made up of two paths, an encoding path and a decoding path. During the encoding also known as contraction, the spatial information is reduced while feature information is increased. The decoding path is also known as the expansive pathway. Features are combined with the spatial information through a sequence of up-convolutions and concatenations with the high-resolution features from the encoding path (Figure 2). The output of the U-Net model is used to generate the dataset for the classification process.[19]

2.4 | Alex-Net

Alex-Net, which was developed in 2021 to classify images for the ImageNet LSVRC-2010 competition, is a deep neural network aimed at mimicking the biological process of human vision. It consists of eight layers which are designed in a way to stand out from any other network architectures.[20]

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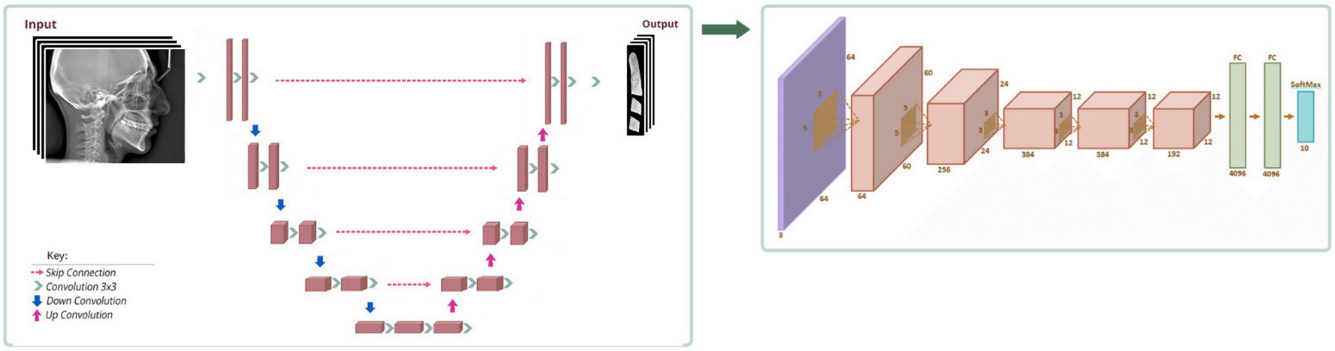


FIGURE 2 The workflow for the segmentation and the classification processes of CVM stage. The first schema represents the U-Net segmentation network. The output of this network feeds the Alex-Net architecture which is used for classification (seen in the right diagram).

2.5 | Implementation of segmentation and classification models

The study algorithm was based on the Python implementation of U-Net for segmentation and Alex-Net for classification.

All training and experiments were done using NVIDIA® GeForce® RTX 2080 Ti GPU.

For segmentation, the following hyperparameters were used: the network was trained using Adam optimizer. The cross-entropy was implemented as a loss function. The batch size was set to 1. The learning rate was set to 0.0002. The model with the best validation loss was chosen for testing.

For classification, the following hyperparameters were used: the network was trained using Adam optimizer. As a loss function cross-entropy was used. The batch size was set to 4. The learning rate was set to 0.0001. The model with the best validation loss was chosen for testing.

2.6 | Statistical analysis

Two weeks after the first labelling and classification stage, 20% of the dataset was reclassified by the same examiner to measure intra-observer reliability and by a second observer in order to measure the agreement level of the measures obtained by the two observers (inter-observer reliability).

Intra-class correlation coefficient (ICC) and Cohen's kappa were conducted using SPSS version 21.0 (SPSS 21.0 Software Package Program Inc.), to measure the intra-and inter-reliability of the observers.

The Confusion matrix, which is also known as the error matrix, is a special table with a layout to visualize the performance and success of the algorithm.[18] This matrix was utilized to assess the achievement of the algorithm.

To quantify the performance of the classification model, the accuracy, specificity, sensitivity metrics and F1 scores were computed.

Accuracy is a metric that is used to describe the performance of a model across all classes and is calculated as the ratio between the number of correct predictions to the total number of predictions.

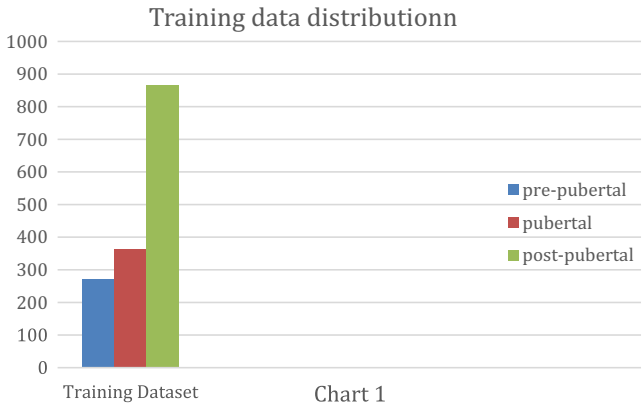


CHART 1 Training data distribution.

Sensitivity (True Positive Rate) refers to the probability of a positive test, conditioned on truly being positive.

Specificity (True Negative Rate) refers to the probability of a negative test, conditioned on truly being negative.

F1-score is an important evaluation metric in machine learning. It efficiently sums up the predictive performance of any model by combining two otherwise competing metrics – precision and recall.

3 | RESULTS

The dataset contained 1501 LCs, the training dataset of U-Net contained 1201 images, and the testing dataset contained 150 images with validation group images which were counted at 150. ICC was 0.973 indicating excellent reliability for the primary observer. The weighted Cohen's kappa standard error was 0.870 ± 0.027. This result indicates high reliability of the observers and proves that there is a good level of agreement between them.

For the classification, the data distribution used in the Alex-Net was as follows: 18.2% in pre-pubertal stage (CS1, CS2), 24.2% in pubertal stage (CS3, CS4) and 57.6% in post-pubertal stage (CS5, CS6; Chart 1).

Sensitivity and specificity mathematically describe the accuracy of a test that reports the presence or absence of a condition.

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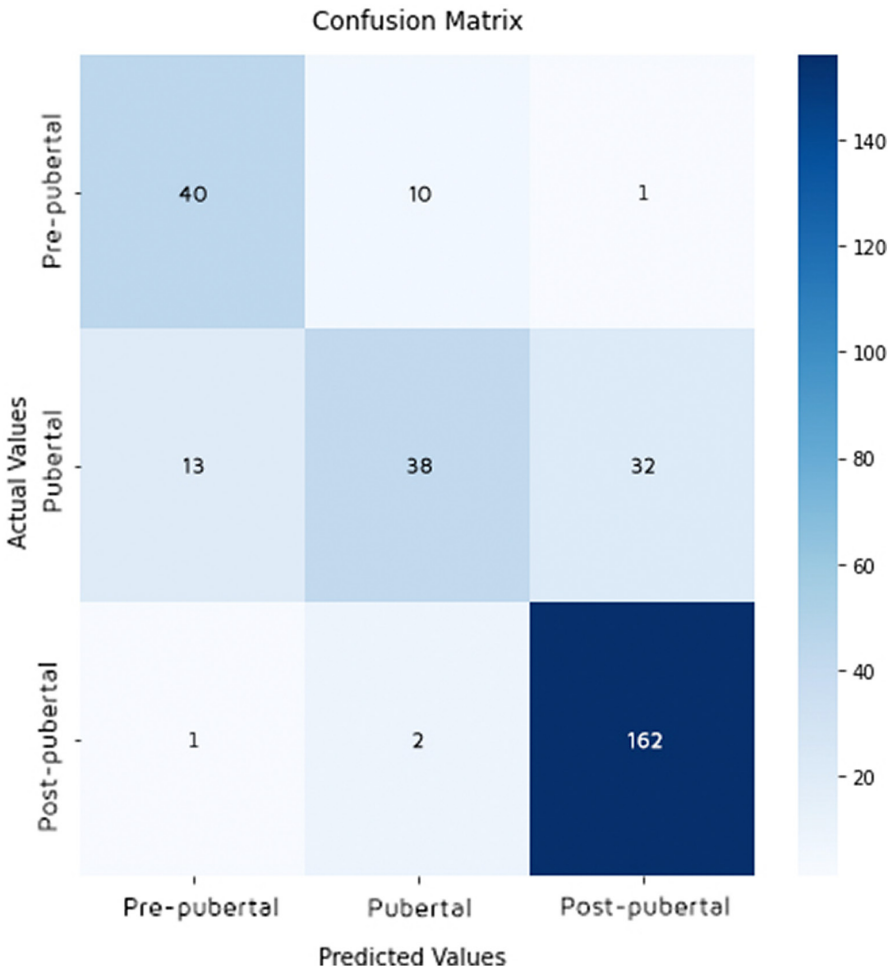


FIGURE 3 Classification model confusion matrix.

TABLE 1 Performance of the algorithm in metric values.

	Accuracy	Sensitivity	Specificity	F1 score
Pre-pubertal	0.9164	0.7843	0.9435	0.7619
Pubertal	0.8094	0.4578	0.9444	0.5714
Post-pubertal	0.8796	0.9818	0.7537	0.9000

Our segmentation network based on U-Net achieved a dice score of 0.931 and an accuracy of 0.998 for the test set. Our classification network based on Alex-Net achieved an overall accuracy of 0.802 for the test dataset. The confusion matrix of the classification model is shown in Figure 3. We achieved per class sensitivity of 0.78, 0.45, 0.98, per class specificity of 0.94, 0.94, 0.75, and in addition, we achieved an F1 score of 0.76, 0.57, 0.90, respectively (Table 1).

4 | DISCUSSION

Multiple factors contribute to the success of orthodontic treatment. Timing is a key factor which can define the success and failure of the treatment, as utilizing the patient growth spurt will guarantee a satisfactory result.

Few indicators were used to determine the growth spurt, i.e., chronological age, sexual maturity, dental age and skeletal age. Furthermore, individual variations were discovered in chronological age and sexual maturity. Evidence from previous studies indicates that dental age is not reliable in assessing skeletal maturity and determining growth spurt.[21]

The CVM staging method remains to be the most implemented one among orthodontists because of its relative ease and the fact that it does not expose the patient to any unnecessary ionizing radiation.

Cervical vertebra maturation stages are determined by the morphological changes that occur to the shape of cervical vertebras. These morphological changes are continuous processes rather than gradual ones which might contribute to some difficulties in determining some CVM stages.

Presently, radiographs can be easily translated into computer language.

Several studies have reported a successful application of AI into cephalometric landmarks detection and tracing.[9,22]

The applications of AI in CVM assessment were pioneered by Amasya et al and Kok et al[15,16,23]

Amasya et al[16] and Kok et al[15] included 647 and 300 LC in their studies, respectively; however, neither study listed the distribution of the dataset. Furthermore, both studies were limited by their sample size and their focus which was solely on the agreement level between AI and the actual stage of CVM.

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Small sample size will cause the training dataset to be insufficient and could potentially cause a scenario known as underfitting, which is when a data model is unable to accurately capture the relationship between the input and output variables.

In an attempt to overcome the sample size limitation described previously, Zhou et al[17] used a sample size of 1080 LC in 2021. Zhou developed an AI algorithm aimed at automatically determining the CVM staging and the data were divided into two groups (n = 980 Training group and n = 100 testing group). Creating the algorithm without a validation set could be acceptable in some cases; however, in an attempt to overcome Zhou's limitation, a larger sample was collected for the present study. (n = 1501) and divided into Training (80%), Validation (10%) and Test groups (10%).

Seo et al[18] evaluated and compared the performance of six state-of-the-art ANN-based deep learning models for CVM staging on LC. Seo is considered the first to implement an unsupervised learning model into the CVM staging process.

Neither Amasya et al[16] nor Kok et al[15] evaluated the AI performance in their studies; however, Zhou did and analysed the results using the F1 score. The results of this study revealed that the AI performance was best at CS6 (F1 score: 85%). Further findings were CS1 (F1 score: 77%) and CS3 proved to have the lowest F1 score at (F1 score:31%).

These findings are in accordance with the results of our study. Our algorithm performed best at pre-pubertal and post-pubertal stages (F1 score 76% and 90%, respectively) and showed (F1 score 57%) in pubertal stages (CS3, CS4) identification which means that the algorithm classified pubertal stage incorrectly in some cases. The reasons for this limitation and for the difference in performance level among the stages could be: (1) insufficient training dataset of pubertal stage, (2) the pubertal stage being short and hard to encounter in clinical practice.[17] Another contributing factor could be (3) the difficulty in determining the shape of C3 and C4 to be either square, rectangular vertical, rectangular horizontal or trapezoidal.

Regarding accuracy, Amasya et al[16] only provided the agreement level among observers and between the observer and the AI algorithm but not the overall accuracy level of the AI model used in the study.

Kok et al[15] provided an accuracy level for each stage (CS1-93%, CS2-89.7%, CS3-68.8%, CS4-55.6%, CS5-47.4%, CS6-78%) that shows low accuracy at CS5; however, that could be because CS5 had the fewest number of samples in the entire study, only 300 LC.

Zhou et al[17] provided the overall accuracy of CVM staging done by AI, which resulted in 71% accuracy. Our AI model proved higher CVM staging accuracy of 80%, which could be because the present study had a larger sample size with a validation dataset to tune the performance of our algorithm.

Seo et al[18] reported accuracy exceeding 90% among the algorithms they tested. An equally distributed sample among all maturation stages could have contributed to the equal accuracy level among the stages of maturation. The sample size of 600 LC is the limitation of the study in question.

A study conducted at the University of Iowa reported that the CVM method shows poor reproducibility among the non-expert examiner; therefore, it would be correct to make the assumption that a non-experienced orthodontist might find it difficult to stage CVM, especially in the pubertal period.[24,25]

There are a limited number of studies on full-automatic CVM determination systems. All of the previously mentioned studies used linear and ratio measurements to determine the changes that occur to the vertebra base and did not deal with the vertebra as a shape. In the present study, our approach was to opt for feeding the vertebras to the AI as individual shapes rather than to use linear measurements, using an unsupervised learning techniques and a large sample size.

Despite the fact that we opted for a larger sample size than any previous study and due to AI being data-hungry, one of the limitations of this study is not having enough training datasets to aid the AI to reach its full potential. One of the other limitations of our study was the fact that the original six stages of maturation according to Baccetti were not applied in its original form in order to limit the effect of the uneven distribution of the training data set.

5 | CONCLUSION

The algorithm which was developed for this study proved the ability to determine the CVM stage with a good level of success. Furthermore, measurement correlation showed almost a perfect agreement between observers which indicates high reliability and reproducibility.

AI-based algorithms can be used for accurate determination of the CVM stage using a Lateral cephalogram in our everyday clinical practice.

AUTHOR CONTRIBUTIONS

M.R. was responsible for collecting the sample, the tracing and labelling process, the classification process, and the writing of the manuscript. Ç.S. was responsible for helping in the tracing and labelling process, also helped in the classification process. N.A. was responsible for the training and testing of the algorithm. L.V. was the supervisor of the research.

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CONFLICT OF INTEREST

The authors have no conflicts of interest to declare that are relevant to the content of this article. The authors did not receive support from any organization for the submitted work.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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